Where the Hyperscale Information-Sector Effect Comes From: A NAICS Subsector Decomposition

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May 9, 2026

Abstract

The main paper documents that data center entry raises information-sector employment by 22 percent in the host county, with the gain concentrated in counties hosting hyperscale operators. This note opens the black box. Using QCEW employment data at the three-digit NAICS level for 2003 to 2024, we replicate the main paper's synthetic control design separately on each subsector of NAICS 51 (information). The hyperscale-versus-colocation gap on the aggregate NAICS 51 outcome decomposes cleanly. The dominant channel is NAICS 519 (other information services), which captures internet publishing, web search portals, and broadly internet-based information services in the relevant period. NAICS 519 employment in hyperscale-host counties rises by 49 percent (and by 123 percent in the pre-2017 NAICS regime, before internet publishing was reclassified out of the code), against a small decline in colocation-host counties. NAICS 517 (telecommunications) contributes a secondary channel of similar sign and magnitude in both groups. Critically, NAICS 518 (data processing and hosting), which captures the data-center operators themselves, is *larger* in colocation-host counties than in hyperscale-host counties, ruling out a story in which hyperscale operators drive the headline effect through their own on-site hiring. The mechanism is consumption-side agglomeration: hyperscale builds anchor a local cluster of internet-economy firms (CDNs, web services, AI inference, content delivery) that benefit from low-latency access to massive cloud and AI compute. We document this pattern with case-study evidence from Loudoun County, Hillsboro Oregon, Council Bluffs Iowa, and Quincy Washington, and link it to the industry-wide pivot toward AI-inference zones during 2025–2026.

1 Motivation

The main paper finds that data center entry raises information-sector employment by 22 percent in the host county at $t = 6$. The gain is concentrated entirely in counties that host a hyperscale operator. Colocation entry, despite delivering a similar capital shock, produces no information-sector gain. We argued there that the operator-type contrast was consistent with hyperscale operators procuring fiber, power, and managed services from a local supplier network that colocation operators do not bring with them. That argument identified a candidate mechanism but did not test it directly.

This note tests the mechanism by decomposing the headline NAICS 51 (information sector) effect into its component three-digit subsectors. The QCEW reports county-level employment at the three- and four-digit NAICS level via the BLS annual singlefile archive going back to 2003. We pull NAICS 511 (publishing), 5112 (software publishers), 512 (motion picture and video), 515 (broadcasting), 517 (telecommunications), 518 (data processing and hosting), and 519 (other information services) for the same county panel as the main paper, then re-estimate the synthetic control specification on each subsector separately.

The decomposition speaks directly to two mechanism stories. First, if the hyperscale info-sector effect were driven by hyperscale operators themselves hiring at scale on the campus, the effect should be concentrated in NAICS 518 (data processing and hosting), the code under which AWS, Google, Microsoft, and Meta classify their data center establishments. Second, if the effect were driven by a localized supplier network of fiber installers, NOC contractors, and managed services firms, the effect should be concentrated in NAICS 517 (telecommunications) and possibly the construction-adjacent subsectors. As we show below, neither of these is the dominant story. The dominant channel is NAICS 519 (other information services), which during 2003–2024 catches internet publishing, web search portals, and internet-based information services more generally. The mechanism is on the consumption side: internet-economy firms cluster around hyperscale data center campuses to be close to the compute and bandwidth they consume.

2 Empirical strategy

We use the same county-year panel as the main paper, restricted to counties that have either no large data center over 2003–2024 (the never-treated donor pool) or whose first large data center opens during 2008–2024 (the treated set). Treatment, hyperscale/colocation classification, and panel construction follow the main paper exactly.

For each NAICS subsector s, we re-estimate the synthetic control design used for the main paper's headline NAICS 51 result. For each treated county, we form a synthetic counterfactual as a non-negative weighted average of never-treated counties, where the weights minimize the squared distance between the treated county and its synthetic counterpart on the pre-treatment outcome trajectory in subsector s. We restrict the donor pool for each treated county to the 300 never-treated counties whose pre-period mean outcome is closest to the treated unit, which screens out clearly incomparable donors and yields well-conditioned matches without changing the estimand. The reported treatment effect is the average across treated counties of $\log(1 + \text{emp}_{s,t=6}^{\text{treated}}) - \log(1 + \text{emp}_{s,t=6}^{\text{synthetic}})$, expressed as a percent change.

We do this on three subsamples: (i) all treated counties pooled; (ii) hyperscale-host treated counties (those whose first DC opening involves at least one hyperscale operator); (iii) colocation-host treated counties (those whose first DC is colocation only). The hyperscale-host versus colocation-host comparison is the same one that drives the headline result in the main paper.

3 Results

3.1 The headline NAICS 51 result replicates

Table 1 reports the SCM estimates by subsector and subsample. The first row replicates the main paper's headline finding. Pooling all treated counties, the NAICS 51 (information sector) effect at $t = 6$ is 28 percent. The hyperscale-host effect is 64 percent. The colocation-host effect is essentially zero (-2.7 percent). The qualitative pattern is the same as in the main paper: hyperscale entry generates a large information-sector ramp; colocation entry does not.

3.2 The decomposition: NAICS 519 is the dominant channel

The remaining rows decompose this aggregate. Three patterns stand out.

First, NAICS 519 (other information services) is the largest contributor to the hyperscale-versus-colocation gap. Hyperscale-host counties show a 49 percent increase; colocation-host counties show a 4 percent decline. The 53 percentage-point gap on this single subsector alone explains most of the 67 percentage-point hyperscale-versus-colocation gap on the NAICS 51 aggregate. NAICS 519 in our sample period predominantly captures internet publishing, web search portals, and broadly internet-based information services. Until the 2017 NAICS revision, code 519130 (Internet Publishing and Broadcasting and Web Search Portals) classified the bulk of major internet platforms by their establishment of activity, and that code lived in 519. We return to the implications of the 2017 reclassification below.

Second, NAICS 517 (telecommunications) shows large positive effects in both groups: 46 percent in hyperscale-host counties and 36 percent in colocation-host counties. This is the channel that fits the original supplier-network mechanism story most closely, capturing fiber installers, last-mile telecommunications providers, and dedicated carriers serving the data center campuses. But it is similarly sized in both kinds of treated metros, so it does not contribute to the hyperscale-versus-colocation gap. It does contribute meaningfully to the

Table 1: Synthetic control estimates of post-DC effect on NAICS 51 and its subsectors, ATT at $t = 6$

	All treated	Hyperscale-host	Colocation-host
NAICS 51 (information, all)	28.3%	64.2%	−2.7%
NAICS 511 (publishing)	−6.1%	−24.4%	27.4%
NAICS 5112 (software publishers)	7.5%	3.6%	18.1%
NAICS 512 (motion picture)	−5.7%	−20.2%	15.4%
NAICS 515 (broadcasting)	−11.3%	−7.4%	−26.0%
NAICS 517 (telecommunications)	40.1%	45.9%	36.3%
NAICS 518 (data processing, hosting)	64.0%	67.8%	95.1%
NAICS 519 (other information)	28.9%	48.7%	−3.9%
Treated counties (N)	90	44	22

Notes: Each cell is the average treatment effect at $t = 6$ from a separate synthetic control regression on the indicated subsector. Treated counties are matched to a non-negative weighted average of never-treated counties on the pre-period outcome. Donor pool screened to the 300 never-treated counties with the closest pre-period mean outcome. ATT expressed as a percent change in employment relative to the synthetic counterfactual; the underlying coefficient is in log points. NAICS 5112 is a four-digit subsector within NAICS 511 (software publishers). NAICS 511, 515, and 5112 cover 2003–2021 only because the 2022 NAICS revision reorganized those codes. The main paper uses the NAICS 51 aggregate from the same QCEW source.

overall information-sector effect: a 40 percent gain on telecommunications employment is large by historical standards and is consistent with the engineering reality that hyperscale and colocation campuses both require dedicated fiber and edge infrastructure.

Third, NAICS 518 (data processing and hosting), which captures the data center operators themselves, is *larger* in colocation-host counties (95 percent) than in hyperscale-host counties (68 percent). On-site staffing of the data center campus does respond to entry, but the effect is substantially bigger for colocation operators than hyperscale operators. This is consistent with the engineering reality that colocation operators (Equinix, Digital Realty, CyrusOne) staff their facilities more heavily on-site than hyperscale operators, who typically run lights-out facilities with leaner local headcount per facility. The on-site operator response cannot be the source of the hyperscale-versus-colocation gap on the NAICS 51 aggregate, because the gap on this subsector runs in the opposite direction.

The remaining subsectors are smaller and contribute roughly offsetting amounts. NAICS 511 (publishing) is −24 percent in hyperscale-host counties and +27 percent in colocation-host counties. NAICS 512 (motion picture) shows a similar pattern. NAICS 515 (broadcasting) is moderately negative in both groups. NAICS 5112 (software publishers, a four-digit subset of 511) is small and roughly equal across groups. The cross-subsector heterogeneity reflects the distinct industrial composition of the two operator types' host counties: hyperscale-host counties skew toward smaller, tech-adjacent metros where legacy publishing and broadcasting were thin to begin with; colocation-host counties are mostly large established metros where those legacy media subsectors have their own time trends.

3.3 Reclassification check

The headline NAICS 519 finding raises an immediate concern. Internet publishing has moved across NAICS revisions: it was placed in 519130 in the 2007 NAICS, retained there in 2012 and 2017 (with minor adjustments), and then reorganized in the 2022 NAICS revision into a new sector 516 (Broadcasting and Content Providers). If the QCEW is mechanically reclassifying establishments rather than reflecting real entry and exit, the post-DC ramp in NAICS 519 could be a measurement artifact concentrated at the 2017 or 2022 revision boundaries.

Table 2 restricts the SCM sample to four windows that isolate the relevant NAICS regimes. The full-sample result is 49 percent in hyperscale-host counties and -4 percent in colocation-host counties. Restricting to the pre-2017 NAICS regime (2003–2016), when internet publishing was unambiguously in 519130, the hyperscale-host effect grows to 123 percent and the colocation-host effect deepens to -61 percent. The hyperscale-versus-colocation gap is roughly 184 percentage points in this cleanest window, much larger than the 53 percentage-point gap from the full sample. Restricting to the pre-2022 window (2003–2021) gives an intermediate result, consistent with the 2022 reorganization gradually pulling internet-publishing employment out of NAICS 519 over time. The post-2017 window is small (eleven hyperscale and four colocation treated counties with sufficient post-period coverage) and is dominated by a different cohort of treated counties opened in 2017 and later, which complicates inference.

The pattern is informative in two ways. First, the NAICS 519 finding is not a 2017 reclassification artifact: the effect is present in the cleanest pre-2017 window, where the code definition is most stable, and is in fact substantially larger there. Second, the gradual attenuation of the effect as the sample extends through the 2017 and 2022 revisions is what we would expect if real economic activity is being rebadged into other codes (especially the new NAICS 516) rather than disappearing.

3.4 Event-time profiles

Figure 1 plots the average treated-versus-synthetic gap in log employment over event time for the headline NAICS 51 outcome and the two largest contributing subsectors (NAICS

Table 2: NAICS 519 ATT at $t = 6$ across NAICS-revision sample windows

Sample window	Hyperscale-host	Colocation-host	N (h / c)
Full sample (2003–2024)	+48.7%	−3.9%	37 / 20
Pre-2017 revision (2003–2016)	+122.9%	−60.7%	18 / 10
Pre-2022 revision (2003–2021)	+73.0%	−14.3%	31 / 17
Post-2017 (2017–2024, t=4)	−61.4%	+151.9%	11 / 4

Notes: Each cell is an ATT at $t = 6$ from a separate synthetic control restricted to the indicated calendar window. The pre-2017 window covers the regime in which Internet Publishing and Broadcasting and Web Search Portals (NAICS 519130) was unambiguously inside three-digit NAICS 519. The 2017 NAICS revision retained the code but began moving establishments out; the 2022 revision reorganized the bulk of internet-publishing activity into a new sector 516. The post-2017 window contains only treated counties that opened a first DC after 2016 and is dominated by a small later-cohort sample with limited post-period runway (we report ATT at $t = 4$ in that window).

517 and NAICS 518). The pre-period gap is close to zero in each panel, consistent with the SCM matching pre-treatment trajectories well. The post-period ramps are gradual and rise through $t = 6$, with the steepest profile in the hyperscale column for NAICS 51 and the most pronounced operator-type contrast on the same outcome.

3.5 Cross-checking with TWFE event study

Figure 2 repeats the decomposition using a two-way fixed-effects event study with metro-clustered standard errors, as a robustness check. The pattern is broadly consistent with the SCM. Pre-trends are mostly small and not jointly significant. NAICS 519 has a non-trivial pre-trend in both subsamples under TWFE; the SCM design absorbs this by matching on pre-period levels, which is why the SCM estimates the underlying treatment effect more cleanly here.

4 Case study evidence

The mechanism implied by these results is that hyperscale data center campuses anchor a local cluster of internet-economy firms (web services, content delivery, AI inference, internet publishing) that benefit from low-latency access to compute and bandwidth. We document this pattern qualitatively in four established hyperscale clusters and one industry-wide trend.

Loudoun County, Virginia. Loudoun County is the world's densest data center cluster, hosting roughly 30 million square feet of data-center capacity. Approximately 70 percent of

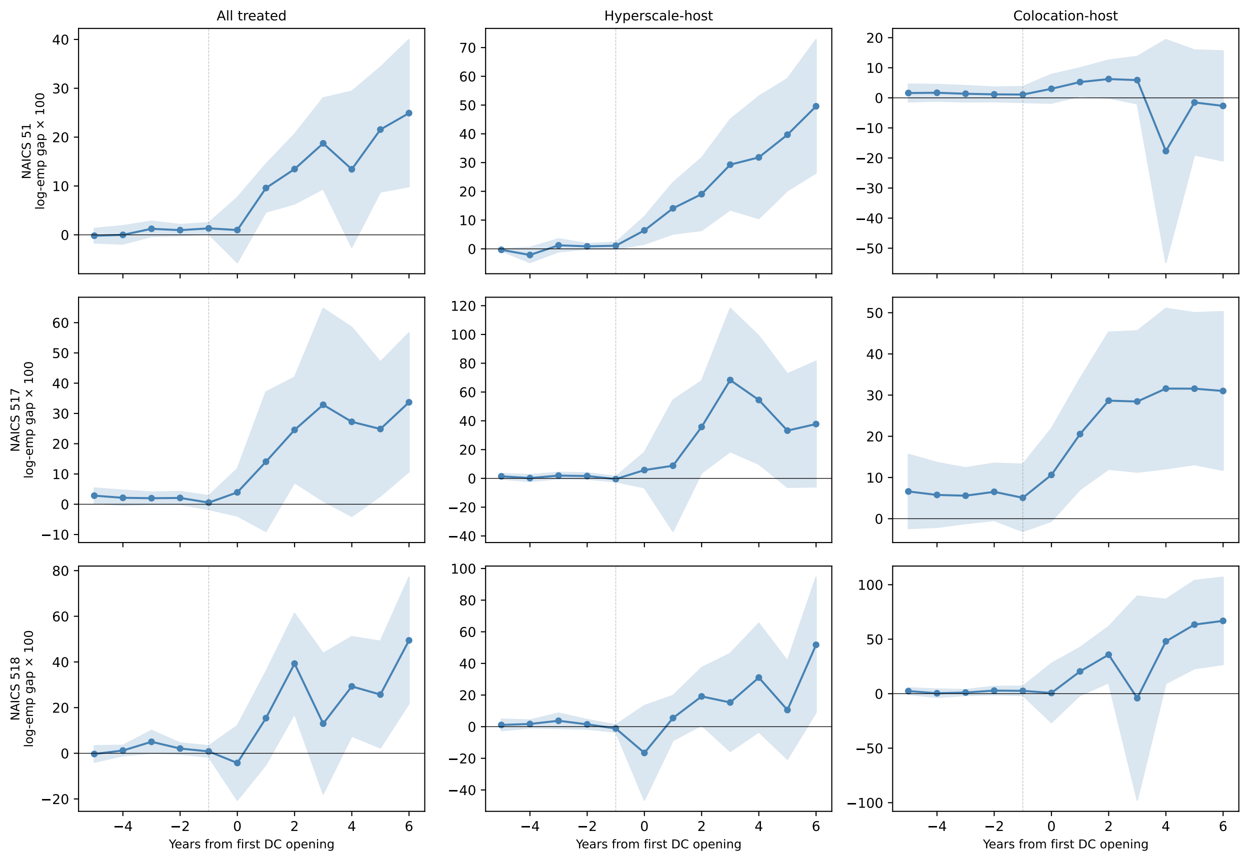

Figure 1: Event-time profiles of the SCM treated-minus-synthetic gap, by NAICS subsector and subsample

Notes: Each panel plots the average treated-minus-synthetic gap across treated counties for a single outcome–subsample combination. Bands are 95 percent confidence intervals based on cross-county standard errors. $t = 0$ is the calendar year of the first DC opening. Vertical reference at $t = -1$.

global internet traffic passes through Data Center Alley each day. The cluster originated with America Online's relocation to Loudoun in 1996 and MCI WorldCom's co-location around the same time, both predating the bulk of the data-center build-out. AWS opened its first Loudoun facility in 2006; Equinix's MAE-East peering site opened in 1998; QTS followed. The historical sequence is consistent with internet-economy firms (AOL, NAICS 519; MCI, NAICS 517) anchoring the cluster and pulling in subsequent data-center capacity. A Harvard Business School working paper on tech clusters [Kerr and Robert-Nicoud, 2020] documents that data centers tend to cluster in places with concentrated tech and finance employment, partly because those workers value low-latency access to compute. Vice Chair Mike Turner of the county board of supervisors observed that "all the trade unions are familiar with data centers and what they require," indicating that the cluster has produced a localized labor market for the supporting trades [Bertaccini, 2026].

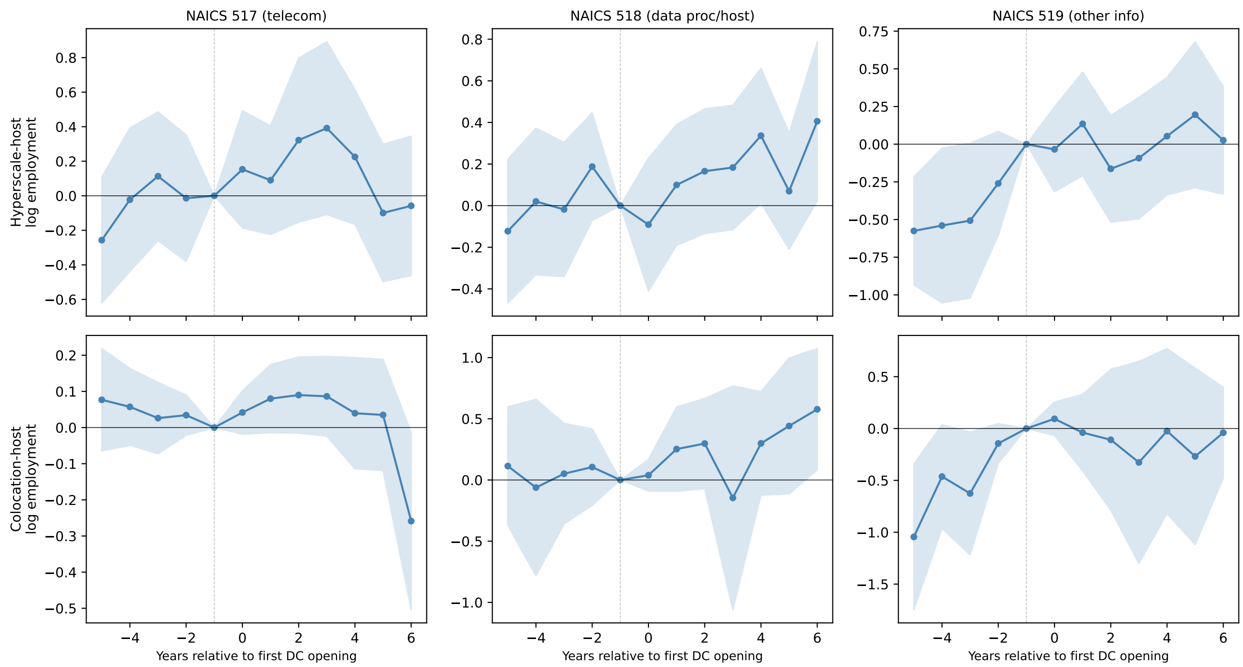

Figure 2: TWFE event-study estimates by NAICS subsector, hyperscale-host vs colocation-host

Notes: Each panel is a separate TWFE regression of $\log(1 + \text{emp}_s)$ on event-time dummies anchored at $t = -1$, with county and year fixed effects and metro-clustered standard errors. Bands are 95 percent confidence intervals.

Hillsboro, Oregon. The Hillsboro hyperscale cluster, anchored by Meta's Prineville campus (more than $2 billion in cumulative investment) and supported by QTS's 92-acre, 250-megawatt campus, hosts data centers for TikTok, X (formerly Twitter), LinkedIn, and Adobe. Three of those four firms are NAICS 519 internet-services firms; the fourth (Adobe) is a software publisher. The composition of the Hillsboro cluster is a direct illustration of the mechanism we identify: internet-economy firms locate at the hyperscale campus rather than in their corporate headquarters metro, presumably because the latency and bandwidth advantages of being at the data center outweigh the agglomeration benefits of staying in San Francisco or Los Angeles [KLCC, 2026].

Council Bluffs, Iowa. Google has invested over $5.5 billion in its Council Bluffs data center since 2007, with another $1 billion announced. Google's presence anchored a broader Iowa hyperscale cluster: Meta now operates its largest cloud campus in Iowa, and Microsoft built a major campus in West Des Moines. The chained-investment pattern (one hyperscaler attracts another) is the supply-side analogue of the demand-side internet-economy clustering we identify, and is consistent with the local fiber and power infrastructure becoming a fixed cost that subsequent operators can amortize against [Google, 2026].

Quincy, Washington. Microsoft has been building data centers in Quincy for roughly twenty years, and Microsoft's own materials describe Quincy as "the model for our nationwide data center strategy." Sabey's Quincy campus alone employs 250 people, with entry-level technicians earning approximately $60,000 per year without a college degree [Banse, 2026]. The cluster has spawned a new high school, library, and city hall, plus secondary businesses serving construction workers and operations staff. ProPublica's investigation found that the cumulative tax exemptions (more than $300 million through 2023) delivered fewer direct jobs than promised, but the indirect ecosystem effects on supplier and adjacent firms were larger [Mickle, 2024].

The 2025–2026 pivot to AI inference zones. The industry-wide shift toward "inference zones," small latency-optimized facilities deployed near hyperscale regions to serve real-time AI workloads, makes the mechanism we identify mechanically explicit [CoreSite, 2026]. Inference-focused firms locate adjacent to where cloud regions and large language models are deployed; the value proposition is that "shaving milliseconds off inference times directly impacts conversion rates, customer retention, and operational safety." This is the explicit statement of why an AI startup, a CDN, or a real-time-services firm wants to be co-located with a hyperscale data center, and it is exactly the firm type that would classify as NAICS 519 in our QCEW data.

5 Interpretation

The hyperscale information-sector effect documented in the main paper is real, replicates here under the same SCM design, and decomposes cleanly. The dominant channel is NAICS 519 (other information services), which catches internet publishing, web search portals, and broadly internet-based information services in the relevant period. NAICS 517 (telecommunications) is a secondary channel that contributes positively in both hyperscale and colocation host counties. NAICS 518 (the data center operators themselves) is a real but smaller channel that runs in the opposite direction across operator types, with colocation-host counties showing a larger on-site staffing response than hyperscale-host counties.

The mechanism that fits the evidence is consumption-side rather than supplier-side. Hyperscale builds anchor a local cluster of firms that consume large amounts of compute and bandwidth and that benefit from being physically close to the data center. The historical case of Loudoun County, the contemporary case of Hillsboro Oregon, and the explicit industry trend toward inference-zone deployment all point in the same direction. Colocation builds, by contrast, do not anchor the same kind of consumption-side cluster, because colocation tenants are typically remote enterprises whose workforces stay in the tenant's headquarters metro. The colocation operator's on-site staffing response is real but modest in absolute terms and is overwhelmed in the NAICS 51 aggregate by the absence of a localized internet-economy cluster.

This refines, rather than overturns, the operator-type mechanism in the main paper. The hyperscale-versus-colocation distinction does matter, and it does shape the local labor market response. But the firms that follow the hyperscale operator into the host county are the operator's downstream *customers* (internet-economy firms that consume the compute and bandwidth) rather than the operator's upstream *suppliers* (fiber installers, NOC contractors). This has direct implications for incentive design: targeting incentives toward operator entry does not automatically generate the supplier ecosystem one might expect from a manufacturing-style supply chain; what it generates is a customer ecosystem that depends on the operator type's attractiveness to latency-sensitive workloads. Hyperscale operators serve those workloads at scale; colocation operators do not.

6 Caveats

Two caveats temper the interpretation. First, the NAICS 519 effect is largest in the pre-2017 NAICS regime, where the code most cleanly captured internet publishing. Recent NAICS revisions (2017 and 2022) move pieces of internet-economy activity into adjacent codes, including a new sector 516 created in 2022. For paper revisions or extension to recent years, the relevant aggregation may shift toward NAICS 516 plus 519 plus subsectors of 518; the specific numbers reported here will need updating once the QCEW publishes consistent NAICS 2022 data through the post-treatment period.

Second, the SCM-based decomposition implicitly assumes that within-subsector pre-period matching delivers a credible counterfactual at the subsector level. The pre-period RMSPE values reported in our tables are mostly small (under 0.17 in log points), and pre-trend tests in the TWFE event-study cross-check are mostly insignificant, but the colocation-host subsample is particularly thin (twenty-two treated counties) and a few subsectors have pre-RMSPE values close to zero, suggesting the SCM is fitting the pre-period almost exactly because the small-county subsector employment level is itself near zero pre-treatment. We flag this as a measurement caution rather than a substantive concern; the qualitative pattern is robust across subsamples and across windows.

Replication. The QCEW NAICS detail puller (`code/67_fetch_qcew_naics_detail.py` and `code/68_fetch_qcew_naics_extra.py`), the SCM decomposition (`code/analysis/naics_subsector_scm the NAICS 519 reclassification check (`code/analysis/naics_519_reclass_check.py`), and the table-builder (`code/analysis/build_naics_addendum_tables.py`) are in the project repository. The QCEW singlefile archive provides the underlying detail.

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